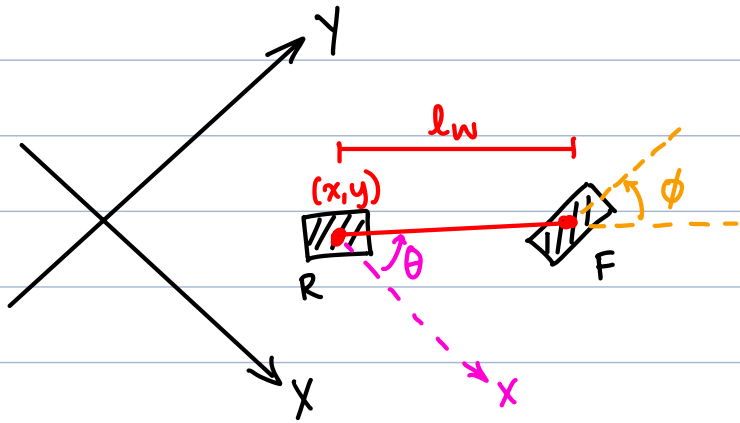


Example: Tricycle/Kinematic Planar Bicycle Model



$$q = (x, y, \theta, \phi)$$

$$Q = \mathbb{R}^2 \times \mathbb{S}^1 \times \mathbb{S}^1 \quad \uparrow \text{ or maybe } [-\pi/3, \pi/3]$$

For rear wheel:

$$\omega_R(q) \dot{q} = \underbrace{\begin{bmatrix} -\sin\theta & \cos\theta & 0 & 0 \end{bmatrix}}_{\omega_R} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix} = 0$$

ω_R : one wheel constraint on rear wheel

For front wheel:

Let (x_F, y_F, θ_F) be the pose of the front wheel

$$\rightarrow \text{i.e. } x_F = x + l_w \cos\theta$$

$$y_F = y + l_w \sin\theta$$

$$\theta_F = \theta + \phi$$

$$\omega_F(q) \dot{q} = 0$$

$$\dot{x}_F = \dot{x} - l_w \dot{\theta} \sin\theta$$

$$\begin{bmatrix} -\sin\theta_F & \cos\theta_F & 0 \end{bmatrix} \dot{q}_F = 0$$

$$\dot{y}_F = \dot{y} + l_w \dot{\theta} \cos\theta$$

$$\dot{\theta}_F = \dot{\theta} + \dot{\phi}$$

$$\begin{bmatrix} -\sin(\theta+\phi) & \cos(\theta+\phi) & 0 \end{bmatrix} \begin{bmatrix} \dot{x} - l_w \dot{\theta} \sin\theta \\ \dot{y} + l_w \dot{\theta} \cos\theta \\ \dot{\theta} + \dot{\phi} \end{bmatrix}$$

$$0 = \begin{bmatrix} -(\dot{x} - l_w \dot{\theta} \sin\theta) \sin(\theta+\phi) & (\dot{y} + l_w \dot{\theta} \cos\theta) \cos(\theta+\phi) & 0 \end{bmatrix}$$

$$0 = -\dot{x} \sin(\theta+\phi) + \dot{y} \cos(\theta+\phi) + l_w \dot{\theta} \underbrace{(\sin\theta \sin(\theta+\phi) + \cos\theta \cos(\theta+\phi))}_{\cos\phi}$$

$$\omega_F(q) \dot{q} = \underbrace{\begin{bmatrix} -\sin(\theta+\phi) & \cos(\theta+\phi) & l_w \cos\phi & 0 \end{bmatrix}}_{\omega_F} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix} = 0$$

$$\Omega(q) = \text{span} \{w_R, w_F\} \subset T_q^*Q \quad \leftarrow \text{space we can't move}$$

↑
codistribution

Now we want to find $\Delta = \Omega^\perp$

$$\begin{aligned} \dim \Delta(q) &= \dim Q - \dim \Omega \\ &= 4 - 2 \\ &= 2 \end{aligned}$$

$$\Delta(q) = \text{span} \{g_1(q), g_2(q)\} \subset T_q Q$$

suppose $\rightarrow$ We want 2 inputs s.t. v_1 corresponds to the speed at the rear and v_2 corresponds to the steering rate.

In general: looking for

$$\begin{bmatrix} -\sin\theta & \cos\theta & 0 & 0 \\ -\sin(\theta+\phi) & \cos(\theta+\phi) & l_w \cos\phi & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = 0$$

↑
 g_i

If v_1 is speed of R, then $\overbrace{a = \cos\theta, b = \sin\theta}^{\text{from diagram}} \Rightarrow \begin{bmatrix} \cos\theta \\ \sin\theta \\ c \\ d \end{bmatrix}$

$$\underbrace{-\sin(\theta+\phi)\cos\theta + \cos(\theta+\phi)\sin\theta}_{-\sin\phi} + cl_w \cos\phi = 0$$

$$\Rightarrow c = \frac{1}{l_w} \tan\phi$$

d arbitrary for now but

NOTE: want $v_2 = \dot{\phi} \rightarrow g_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

$$g_1 = \begin{bmatrix} \cos\theta \\ \sin\theta \\ \frac{1}{l_w} \tan\phi \\ 0 \end{bmatrix}$$

$$\therefore G(q) = \begin{bmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ \frac{1}{l_w} \tan\phi & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

two directions we can move $\ddot{}$